

Project: Heartattack Risk Prediction(Using synthetic data)

Importing Libraries

```
In [7]: import pandas as pd
import numpy as np
```

Initialize Random Seed and Dataset Size

```
In [8]: np.random.seed(43)
patients=10000
```

Generate Synthetic Patient Data

```
In [9]: data={
    "Age" : np.random.randint(14,70, patients),
    "Gender" : np.random.choice(["Male", "Female"], patients),
    "Marital_Status" : np.random.choice(["Married", "Single"], patients),
    "Previous_Heart_Disease" : np.random.choice(["Yes", "No"], patients),
    "Family_History_Heart_Disease" : np.random.choice(["Yes", "No"], patients),
    "History_of_Heart_Attack" : np.random.choice(["Yes", "No"], patients),
    "Bloodpressure" : np.random.choice(["LowBP", "HighBP"], patients),
    "Diabetes" : np.random.choice(["Yes", "No"], patients),
    "Kidney_Disease" : np.random.choice(["Yes", "No"], patients),
    "Obesity" : np.random.choice(["Yes", "No"], patients),
    "Thyroid_Disorders" : np.random.choice(["Yes", "No"], patients),
    "OtherHeart_Conditions" : np.random.choice(["Yes", "No"], patients),
    "Weight" : np.clip(np.random.normal(75,15,patients), 20,100).round(1),
    "Height" : np.clip(np.random.normal(75,15,patients), 1,10).round(1),
    "Hemoglobin": np.clip(np.random.normal(75,15,patients), 20,100).round(1),
    "exercise_days": np.random.randint(0, 8, patients),
    "Fruit_and_Vegetable_servings": np.random.randint(0, 11, patients),
    "Fast_Food_Consumption": np.random.randint(0, 8, patients),# Times per week
    "Number_of_Cigarettes_per_Day": np.random.randint(0, 41, patients),
    "Alcohol_Consumption" : np.random.choice(["high", "low", "moderate"], patients),
    "Anxiety_Level": np.random.randint(1, 11, patients),
    "Sleep_Duration": np.round(np.random.uniform(3, 12, patients),1),
    "Air_Pollution_Exposure": np.random.choice(["Low", "High", "Moderate", "Very High"], patients),
    "Urban_vs_Rural_Residence": np.random.choice(["Yes", "No"], patients),
    "Access_to_Healthcare_Facilities" : np.random.choice(["Poor", "Good", "Average"], patients),
    "Healthcare_Insurance_Status": np.random.choice(["Yes", "No"], patients),
    "Occupational_Stress": np.random.choice(["Yes", "No"], patients),
    "BMI" : np.round(np.random.uniform(16, 45, patients),1)
}
```

Creating the DataFrame

```
In [10]: df = pd.DataFrame(data)
df
```

```
Out[10]:
```

	Age	Gender	Marital_Status	Previous_Heart_Disease	Family_History_Heart Disease	Histo Disease
0	18	Feamle	Single	Yes		No
1	14	Male	Married	Yes		Yes
2	63	Feamle	Single	No		Yes
3	35	Male	Single	Yes		No
4	30	Male	Single	Yes		No
...	...	...	...	...		...
9995	14	Feamle	Married	No		Yes
9996	45	Feamle	Married	Yes		No
9997	43	Male	Single	No		Yes
9998	23	Male	Married	Yes		No
9999	38	Male	Single	Yes		No

10000 rows × 28 columns



Calculating Heart Disease Risk Score

```
In [12]: risk_score = (

    # Age
    0.10 * (df["Age"] >= 55).astype(int) +

    # Medical History
    0.20 * (df["Previous_Heart_Disease"] == "Yes").astype(int) +
    0.15 * (df["History_of_Heart Attack"] == "Yes").astype(int) +
    0.10 * (df["Family_History_Heart Disease"] == "Yes").astype(int) +

    # Chronic Diseases
    0.10 * (df["Diabetes"] == "Yes").astype(int) +
    0.08 * (df["Kidney_Disease"] == "Yes").astype(int) +
    0.08 * (df["Obesity"] == "Yes").astype(int) +

    # Blood Pressure
    0.10 * (df["Bloodpressure"] == "HighBP").astype(int) +

    # Lifestyle
    0.08 * (df["exercise_days"] <= 1).astype(int) +
    0.05 * (df["Fast_Food_Consumption"] >= 4).astype(int) +
    0.05 * (df["Fruit_and_Vegetable_servings"] <= 2).astype(int) +

    # Smoking
    0.10 * (df["Number_of_Cigarettes_per_Day"] >= 10).astype(int) +
```

```

# Alcohol
0.05 * (df["Alcohol_Consumption"] == "high").astype(int) +

# Stress & Mental Health
0.05 * (df["Anxiety_Level"] >= 8).astype(int) +
0.05 * (df["Occupational_Stress"] == "Yes").astype(int) +

# Sleep
0.05 * (df["Sleep_Duration"] < 6).astype(int) +

# Environment
0.03 * (df["Air_Pollution_Exposure"].isin(["High", "Very High"])).astype(int)

# BMI
0.08 * (df["BMI"] >= 30).astype(int) -

# Protective Factors
0.05 * (df["exercise_days"] >= 5).astype(int) -
0.05 * (df["Fruit_and_Vegetable_servings"] >= 5).astype(int) -
0.03 * (df["Access_to_Healthcare_Facilities"].isin(["Good", "Verygood"])).as

)

risk_score += np.random.normal(0, 0.03, patients)

df["Heart_Attack_Risk"] = (risk_score >= 0.45).astype(int)

df["Heart_Attack_Risk"]

```

```

Out[12]: 0      1
         1      1
         2      1
         3      1
         4      1
         ..
        9995     1
        9996     1
        9997     1
        9998     1
        9999     1
        Name: Heart_Attack_Risk, Length: 10000, dtype: int64

```

Saving into CSV File

```

In [13]: df.to_csv("Heart_Attack_Risk_synthetic_data", index = False)

```

```

In [ ]:

```